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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
16/337,906	03/28/2019	Steffen Gunther Wiesner	1141/US/2	1008

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EXAMINER
KAZIMI, MAHMOUD M

ART UNIT	PAPER NUMBER
3662	

NOTIFICATION DATE	DELIVERY MODE
02/27/2024	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 16/337,906	Applicant(s) Wiesner et al.	
	Examiner MAHMOUD M KAZIMI	Art Unit 3662	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 04/28/2023.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☑ This action is **FINAL.** 2b) ☐ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-5,7-10,12,14-16 and 18-19 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-5,7-10,12,14-16 and 18-19 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☐ The drawing(s) filed on _____ is/are: a)☐ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☐ Notice of References Cited (PTO-892)

2) ☐ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

Notice of Pre-AIA or AIA Status

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Status of Claims

This communication is in response to applicant's filing dated 12/30/2021. Claims 6, 11, 13 and 17 are canceled. Claims 1-5, 7-10, 12, 14-16 and 18 are currently pending.

Response to Argument

Examiner withdraws the rejections of claims 1-5, 7-10, 12, 14-16 and 18 under 35 USC § 103. However, the rejection under 35 USC § 101 directed to non-statutory subject matter of claims 1-5, 7-10, 12, 14-16 and 18 is maintained. The rejections are as stated below.

With respect to the previous 35 U.S.C. 101 rejection of claim 1, the proposed amendments do not overcome the 35 U.S.C. 101 rejection. Applicant amended the claim to include "...generate an aggregate profile that represents a time dependence of the relative value of the parking related parameter for that group, wherein the aggregate profile is used as a representation of the time dependence of the parking related parameter for all the segments of that group, generating a parking search route based on one or more of the aggregate profiles, the parking search route including a set of instructions indicative of the parking search route and outputting instructions from the

set of instructions for guiding a user on the parking search route as the user travels the parking search route..." The concept is still directed to "**a method for generating parking related data**," which does not make it less abstract.

The same updated analysis based on the new 2019 Patent Eligibility Guidance (2019 PEG) applies to the newly added claimed limitations as discussed in the previous office action.

As a result, **Step 2A Prong 1** determines if a claim is directed to those grouping and subgroupings along with an explanation of why it is directed to such.

> "First, the rejection should identify the judicial exception (i.e., abstract idea enumerated in Section I of the 2019 PEG, laws of nature, or a natural phenomenon) by referring to what is recited (i.e., set forth or described) in the claim and explaining why it is considered to be an exception (Step 2A Prong One). There is no requirement for the examiner to provide further support, such as publications or an affidavit or declaration under 37 CFR 1.104(d)(2), for the conclusion that a claim recites a judicial exception."
>
> "For abstract ideas, the rejection should explain why a specific limitation(s) recited in the claim falls within one of the enumerated groupings of abstract ideas (i.e., mathematical concepts, mental processes, or certain methods of organizing human activity) or provide a justification for why a specific limitation(s) recited in the claim is being treated as an abstract idea if it does not fall within the enumerated groupings of abstract ideas in accordance with the "tentative abstract idea" procedure in the 2019 PEG."

In the Non-Final mailed 08/04/2022 examiner performs the analysis and clarifies that "the abstract idea noted in the independent claims...are directed to a "Mental Processes." Hence, examiner has indicated that these identified limitations are directed

to "a method for generating parking related data" and has provided a justification for why these limitations fall within one of the enumerated groupings of abstract ideas (i.e. concepts performed in the human mind). This is sufficient under the guidelines of the 2019 PEG and October 2019 Update as cited above. Accordingly, it seems reasonable for the examiner to group the abstract idea under "Mental processes." as enumerated in Section I of the 2019 PEG.

Prong Two:

With respect to **Step 2A, prong two,** Integration into a practical application requires an additional element(s) or a combination of additional elements in the claim to apply, rely on, or use the judicial exception in a manner that imposes a meaningful limit on the judicial exception, such that the claim is more than a drafting effort designed to monopolize the exception. Limitations that are not indicative of integration into a practical application are those that are mere instructions to implement an abstract idea on a computer, or merely uses a computer as a tool to perform an abstract idea.-see MPEP 2106.05(f).

In contrast, the instant claims are different, the focus of the claims is not on such an improvement in computers as tools, but on certain independently abstract ideas that use computers as tools. The claims here are not directed to a specific improvement to computer functionality nor an inventive solution to any computer specific problem.

Examiner submits that under the current 35 USC 101 examining practice, the existence of such novel features would still not cure the deficiencies with respect to the abstract idea. See for example: *Ultramercial, Inc. v. Hulu, LLC, 112 USPQ2d 1750, U.S. Court of Appeals Federal Circuit, No. 2010-1544, Decided November 14, 2014, 2014 BL*

320546, 772 F.3d 709, Page 1754 last two ¶ : Indeed, in this in instant case, the limitations simply narrow or limit the abstract idea without providing anything significantly more than the abstract idea itself.

Lastly, dependent claims do not include additional elements that are sufficient to amount to significantly more than the judicial exception because the additional elements are simply steps performed by a generic computer. The claim merely amounts to the application or instructions to apply the abstract idea on a processor, and is considered to amount to nothing more than requiring a generic processor to merely carry out the abstract idea itself.

For these reasons the rejection under 35 U.S.C. § 101 directed to non-statutory subject matter set forth in this office action is maintained.

Claim Rejections - 35 USC § 101

7. 35 U.S.C. 101 reads as follows:

> Whoever invents or discovers any new and useful process, machine, manufacture, or composition of matter, or any new and useful improvement thereof, may obtain a patent therefor, subject to the conditions and requirements of this title.

8. **Claims 1-5, 7-10, 12, 14-16, 18 and 19** are rejected under 35 U.S.C. 101 because the claimed invention is directed to an abstract idea without significantly more.

Under Step 2A – Prong 1:

Claim 1, 16 and 18 recite the abstract idea concept of a method (claim 1), system (claim 16), and non-transitory computer readable medium (claim 18) for generating parking related data within a geographic network. This abstract idea is described in at least claims 1, 16 and 18 by, **analysing, for each one of the plurality of trips, the**

positional data to identify a parking space search portion of the trip; analysing the positional data corresponding to the parking search portions of the trips to obtain, for each one of at least one set of a plurality of different time periods, a relative value of the parking related parameter associated with each one of a plurality of segments of the map in respect of the time period, the parking related parameter being a time dependent parameter; clustering, for the or each set of time periods, the segments into groups of segments based on the relative values of the parking related parameter obtained for the segments in each time period of the set, wherein, for each of the time periods of the set of time periods, the segments in each group of segments are considered to have similar relative values of the parking related parameter; and using, for each group, relative parking related parameter data associated with each of the segments of the group to obtain an aggregate profile indicative of the time dependence of the relative value of the parking related parameter, the aggregate profile being applicable to each of the segments of the group, are considered mental process steps. The identified claim limitations that recites an abstract idea fall within the enumerated groupings of abstract ideas in Section 1 of the 2019 Revised Patent Subject Matter Eligibility Guidance published in the Federal Register (84 FR 50) on January 7, 2019.

The limitations of **analysing, for each one of the plurality of trips, the positional data**, and **analysing the positional data corresponding to the parking space, clustering, the segments into groups of segments based on the relative values of the parking related parameter obtained for the segments in each time period of the set, using, the aggregate profile being applicable to each of the**

segments of the group as drafted, are process that, under its broadest reasonable interpretation, covers performance of the limitation in the mind but for the recitation of generic computer components. That is, other than reciting "A method for generating parking related data" nothing in the claim element precludes the step from practically being performed in the mind. If a claim limitation, under its broadest reasonable interpretation, covers performance of the limitation in the mind but for the recitation of generic computer components, then it falls within the "Mental Processes" grouping of abstract ideas. Accordingly, the claim 1 recites an abstract idea.

Under Step 2A – Prong 2:

The claims recite elements additional to the abstract idea. However, these additional elements fail to integrate into a practical application.

Claim 1 recites, **obtaining positional data relating to a plurality of trips made by vehicles in the navigable network, each trip being a journey between stationary periods indicative of the vehicle being in a parked state and electronic map** and **navigable network and electronic map, generate an aggregate profile that represents a time dependence of the relative value of the parking related parameter for that group, wherein the aggregate profile is used as a representation of the time dependence of the parking related parameter for all the segments of that group, generating a parking search route based on one or more of the aggregate profiles, the parking search route including a set of instructions indicative of the parking search route and outputting instructions from the set of instructions for guiding a user on the parking search route as the user travels the parking search route.** Which is mere data gathering that is simply employed as a tool

to collect information and display results, which is insignificant extra solution activity as the step simply gathers data necessary to perform the abstract idea. These additional steps amount necessary data gathering and generic computer components wherein all uses of the recited abstract idea require such data gathering or data output. See MPEP 2106.05(g).

Under Step 2B:

Regarding Step 2B of the 2019 PEG, independent claim 1 does not include additional elements (considered both individually or in combination) that are sufficient to amount to significantly more than the judicial exception for the same reasons to those discussed above with respect to determining that the claim does not integrate the abstract idea into a practical application. As discussed above with respect to integration of the abstract idea into a practical application, the additional elements amount to nothing more than applying the exception using a generic computer component. Generally applying an exception using a generic computer component cannot provide an inventive concept. And as discussed above, the additional limitations of **obtaining positional data relating to a plurality of trips made by vehicles in the navigable network, each trip being a journey between stationary periods indicative of the vehicle being in a parked state and electronic map** and **navigable network and electronic map**, the examiner submits that these limitations are insignificant extra-solution activities.

Further, a conclusion that an additional element is insignificant extra-solution activity in Step 2A should be re-evaluated in Step 2B to determine if they are more than what is well-understood routine and conventional activity in the field. The additional

limitations of obtaining positional data relating to a plurality of trips made by vehicles in the navigable network, each trip being a journey between stationary periods indicative of the vehicle being in a parked state and electronic map and navigable network and electronic map, are well-understood, routine and conventional activities because the background recites that obtaining steps are done by a GPS/GNSS receiver and an electronic map (or mathematical graph, as it is sometimes known), in its simplest form, is effectively a database containing data representative of nodes, most commonly representative of road intersections, and lines between those nodes representing the roads between those intersections. In more detailed digital maps, lines may be divided into segments defined by a start node and end node" and that the navigable network is simply a road network. Further the mere collection of data or receipt of data over a network is a well-understood, routine and conventional function when it is claimed in a merely generic manner (as it is here). See MPEP 2106.05(d). Therefore, claims 1, 16 and 18 are ineligible under 35 U.S.C. 101.

Dependent claims 2-5, 7-10, 12, 14, 15 and 19 include limitations of claim 1 and only further defines mental process of claim 1. These further limitations are considered mental process without any significantly more elements to the abstract idea. The limitations of dependent claims are directed toward additional aspects of the judicial exception and/or well-understood, routine and conventional additional elements that do not integrate the judicial exception into a practical application. Mere collection of data or receipt of data over a network is well-understood, routine and conventional. See MPEP 2106.05(d).

Claims 1-5, 7-10, 12, 14-16, 18 and 19 are rejected under 35 U.S.C. 101 because the claimed invention is directed to an abstract idea without significantly more.

Allowable Subject Matter

13. **Claims 1-5, 7-10, 12, 14-16, 18 and 19** would be allowable if rewritten or amended to overcome the rejection(s) under 35 U.S.C. 101, set forth in this Office action..

Kandal discloses wherein the parking related parameter is a time dependent parameter, and the method further comprises:

obtaining positional data relating to trips made by vehicles in the navigable network, each trip being a journey between stationary periods indicative of the vehicle being in a parked state (***See at least column 3 lines 7-12***);

analysing, for each one of the plurality of trips, the positional data to identify a parking space search portion of the trip, the analyzing including analysising the respective positional data to identify a start of a parking space search during the trip that defines the start of the parking space search portion of the trip (***See at least column 5 lines 15-18***); and

analysing the positional data corresponding to the parking search portions of the trips to obtain, for each one of at least one set of a plurality of different time periods, a relative value of the parking related parameter associated with each one of a plurality of segments of the electronic map in respect of the time period (***See at least column 4 lines 83-88 and column 5 lines 15-18***).

None of the prior art of record reference taken either together or in combination with the prior art record discloses:

(**Claims 13, 16 and 18**) "clustering, for the or each set of time periods, the segments into groups of segments based on the relative values of the time dependent parameter obtained for the segments in each time period of the set, wherein, for each of the time periods of the set of time periods, the segments in each group of segments are considered to have similar relative values of the parking related parameter; and using, for each group, relative parking related parameter data associated with each of the segments of the group to obtain an aggregate profile indicative of the time dependence of the relative value of the parking related parameter, the aggregate profile being applicable to each of the segments of the group."

In combination with the remaining elements and features of the claimed invention. It is for those reasons that the applicant's invention defines over the prior art of record.

Conclusion

Applicant's amendment necessitated the new ground(s) of rejection presented in this Office action. Accordingly, **THIS ACTION IS MADE FINAL**. See MPEP § 706.07(a). Applicant is reminded of the extension of time policy as set forth in 37 CFR 1.136(a).

A shortened statutory period for reply to this final action is set to expire THREE MONTHS from the mailing date of this action. In the event a first reply is filed within TWO MONTHS of the mailing date of this final action and the advisory action is not

mailed until after the end of the THREE-MONTH shortened statutory period, then the shortened statutory period will expire on the date the advisory action is mailed, and any extension fee pursuant to 37 CFR 1.136(a) will be calculated from the mailing date of the advisory action. In no event, however, will the statutory period for reply expire later than SIX MONTHS from the date of this final action.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to MAHMOUD M KAZIMI whose telephone number is (571)272-3436. The examiner can normally be reached M-F 7am-5pm.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Aniss Chad can be reached on 571-270-3832. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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/M.M.K./
Examiner, Art Unit 3662

/ANISS CHAD/
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